Chapter 7: Multi-way Search Trees

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 7 (Multi-way Search Trees) of Data Structures.

Technical / Interview Questions

1. What is a multi-way search tree? How does it differ from a binary search tree?
2. Why are B-trees preferred for database indexing?
3. Define the order of a B-tree.
4. Explain the properties of a B-tree of order m.
5. What is the minimum and maximum number of keys in a B-tree node?
6. Explain the role of node splitting in B-trees.
7. How does B-tree insertion maintain balance?
8. Describe the deletion process in a B-tree.
9. What is redistribution in B-trees?
10. What is node merging in B-trees?
11. What is the height of a B-tree and why is it important?
12. Explain the B-tree Abstract Data Type (ADT).
13. List the basic operations supported by a B-tree ADT.
14. How does the search operation work in a B-tree?
15. Explain the difference between B-tree and B+ tree.
16. Why are B+ trees used in file systems?
17. Explain the structure of internal and leaf nodes in a B+ tree.
18. What is the advantage of linking leaf nodes in B+ trees?
19. Explain range queries in B+ trees.
20. What is a 2–3 tree?
21. What are the types of nodes in a 2–3 tree?
22. Why are 2–3 trees always height-balanced?
23. Explain insertion in a 2–3 tree.
24. Explain deletion in a 2–3 tree.
25. What is a 2–3–4 tree?
26. How is a 2–3–4 tree related to Red–Black trees?
27. What is the difference between a 3-node and a 4-node?
28. Explain node splitting in a 2–3–4 tree.
29. What is a B* tree?
30. How does a B* tree improve space utilization?
31. Explain redistribution in B* trees.
32. Compare B-tree, B+ tree, and B* tree.
33. What is the time complexity of search, insert, and delete operations in B-trees?
34. Explain why multi-way trees reduce disk I/O.
35. Discuss applications of B-trees in database management systems.

Programming Exercises

1. Write a program to implement a B-tree node structure.

2. Write a function to create a B-tree.
3. Implement the search operation in a B-tree.
4. Write a program to insert a key into a B-tree.
5. Implement node splitting during B-tree insertion.
6. Write a function to traverse a B-tree.
7. Implement deletion in a B-tree.
8. Write a program to perform redistribution in a B-tree.
9. Implement node merging in a B-tree.
10. Write a program to calculate the height of a B-tree.
11. Implement a 2–3 tree node structure.
12. Write a function to insert elements into a 2–3 tree.
13. Implement search in a 2–3 tree.
14. Write a program to traverse a 2–3 tree.
15. Implement insertion in a 2–3–4 tree.
16. Write a program to perform node splitting in a 2–3–4 tree.
17. Implement deletion in a 2–3–4 tree.
18. Write a program to simulate a B+ tree leaf node.
19. Implement search in a B+ tree.
20. Write a program to insert a key in a B+ tree.
21. Implement range query operation in a B+ tree.
22. Write a function to link leaf nodes in a B+ tree.
23. Implement node splitting in a B+ tree.
24. Write a program to simulate B* tree node redistribution.
25. Implement insertion in a B* tree.
26. Write a program to calculate node utilization in B* trees.
27. Implement traversal of a B* tree.
28. Write a program to compare B-tree and B+ tree performance.
29. Write a program to display all keys within a given range using a B+ tree.
30. Implement a menu-driven program for B-tree operations.

Multiple Choice Questions

1. A B-tree is a

(a) Binary tree
(b) Balanced multi-way search tree
(c) Linear structure
(d) Heap

2. In a B-tree of order m, the maximum number of children is

(a) m−1
(b) m
(c) m+1
(d) 2m

3. Minimum number of children of a non-root B-tree node is

(a) m/2
(b) [m/2]
(c) m−1
(d) 2

4. All leaves in a B-tree appear at

(a) Different levels
(b) Same level
(c) Random levels
(d) Root level

5. B-trees are mainly used in

(a) Sorting
(b) Database indexing
(c) Graph traversal
(d) Memory allocation

6. In B+ trees, actual data is stored in

(a) Internal nodes
(b) Root node
(c) Leaf nodes
(d) All nodes

7. Leaf nodes in B+ trees are

(a) Not connected
(b) Connected sequentially
(c) Connected randomly
(d) Connected to root

8. A node with two keys in a 2–3 tree is called

(a) 2-node
(b) 3-node
(c) 4-node
(d) Leaf node

9. A 4-node appears in

(a) B-tree
(b) B+ tree
(c) 2–3–4 tree
(d) AVL tree

10. In B-trees, node overflow is resolved by

(a) Rotation
(b) Splitting
(c) Deletion
(d) Traversal

11. B* trees require nodes to be at least

(a) 1/2 full
(b) 2/3 full
(c) 3/4 full
(d) Completely full

12. The time complexity of search in a B-tree is

(a) O(n)
(b) O(log n)
(c) $O(n^2)$
(d) O(1)

13. A B-tree node with k children contains

(a) k keys
(b) $k-1$ keys
(c) $k+1$ keys
(d) k^2 keys

14. Which tree supports efficient range queries?

(a) B-tree
(b) AVL tree
(c) B+ tree
(d) Binary tree

15. Which structure is conceptually equivalent to a Red–Black tree?

(a) B-tree
(b) 2–3 tree
(c) 2–3–4 tree
(d) B+ tree

16. B-trees minimize

(a) CPU usage
(b) Disk access
(c) Memory usage
(d) Network usage

17. Insertion in a B-tree occurs at

(a) Root
(b) Internal nodes
(c) Leaf nodes
(d) Random node

18. Maximum keys in a B-tree node of order m is

(a) m
(b) $m-1$
(c) $m+1$
(d) 2m

19. B+ tree internal nodes store

(a) Data only
(b) Keys only
(c) Data and keys
(d) Pointers only

20. Traversal of B-trees gives

(a) Sorted output
(b) Random output
(c) Reverse order
(d) Heap order

21. A B-tree improves performance by

(a) Increasing height
(b) Reducing height
(c) Removing nodes
(d) Increasing nodes

22. B+ trees are used in

(a) Databases
(b) Graph algorithms
(c) Compression
(d) Sorting

23. In 2–3 trees each node can have

(a) 2 or 3 children
(b) 1 or 2 children
(c) 3 or 4 children
(d) Any number

24. B* trees reduce

(a) Node splitting
(b) Searching
(c) Traversal
(d) Memory

25. A node with one key in a 2–3 tree is

(a) 2-node
(b) 3-node
(c) 4-node
(d) Root node

26. B+ tree leaf nodes store

(a) Data records
(b) Only pointers
(c) Only keys
(d) Parent keys

27. B-trees are best suited for

(a) RAM-based storage
(b) Disk-based storage
(c) Cache memory
(d) Registers

28. B-tree height is typically

(a) Large
(b) Small
(c) Infinite
(d) Random

29. Which tree has linked leaf nodes?

(a) B-tree
(b) B+ tree
(c) AVL tree
(d) Heap

30. B* tree is an improvement of

(a) AVL tree
(b) B-tree
(c) Heap
(d) Trie

31. Minimum children in B-tree root (if not leaf)

(a) 1
(b) 2
(c) 3
(d) m

32. B-tree search starts from

(a) Leaf
(b) Root
(c) Middle node
(d) Random node

33. B+ tree internal nodes contain

(a) Data
(b) Keys and pointers
(c) Only pointers
(d) Records

34. Multi-way trees reduce

(a) Tree height
(b) Keys
(c) Nodes
(d) Traversal

35. In B-trees nodes store

(a) Keys only
(b) Data only
(c) Keys and data
(d) None

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	B	3	B	4	B	5	B
6	C	7	B	8	B	9	C	10	B
11	B	12	B	13	B	14	C	15	C
16	B	17	C	18	B	19	B	20	A
21	B	22	A	23	A	24	A	25	A
26	A	27	B	28	B	29	B	30	B
31	B	32	B	33	B	34	A	35	C